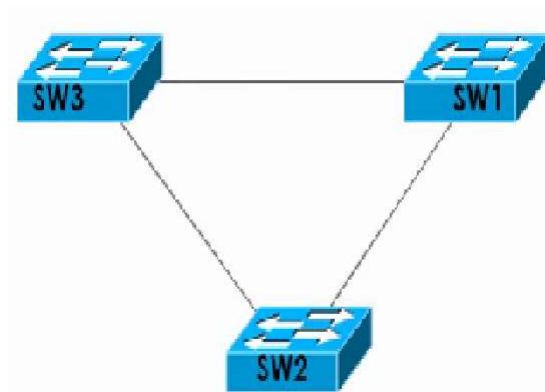


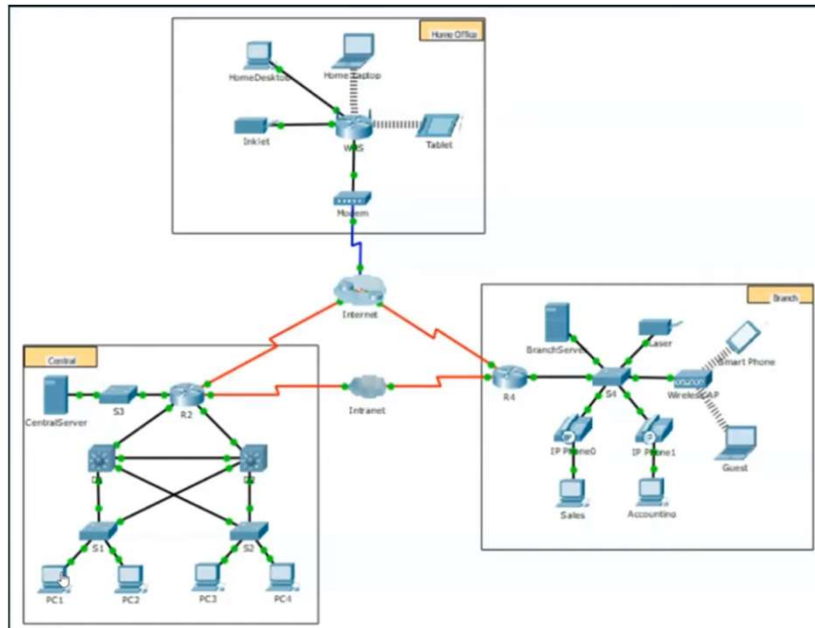
- REDUNDANT LINKS
- Issues (layer2 loop - broadcast storm - Duplicate Unicast frame)
- Spanning tree protocol
- Choose the root bridge
- Select root ports

1

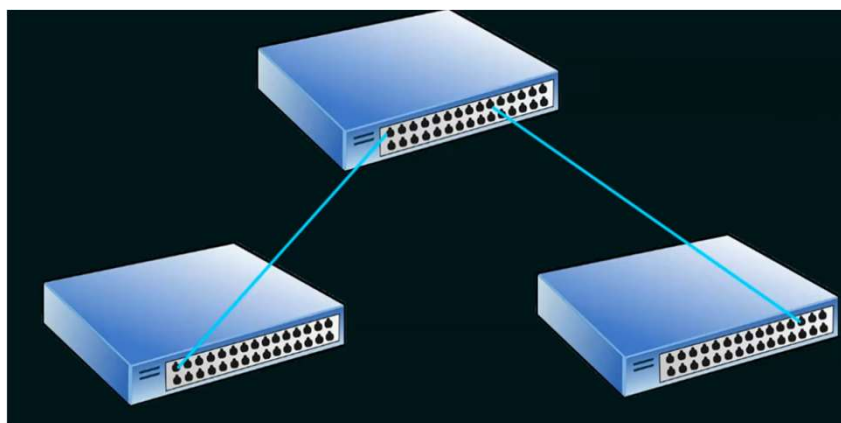
REDUNDANT LINKS



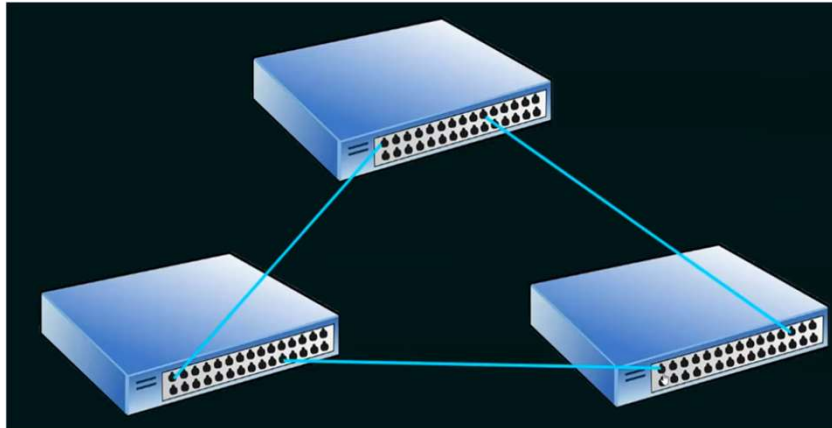
2



3



4



5

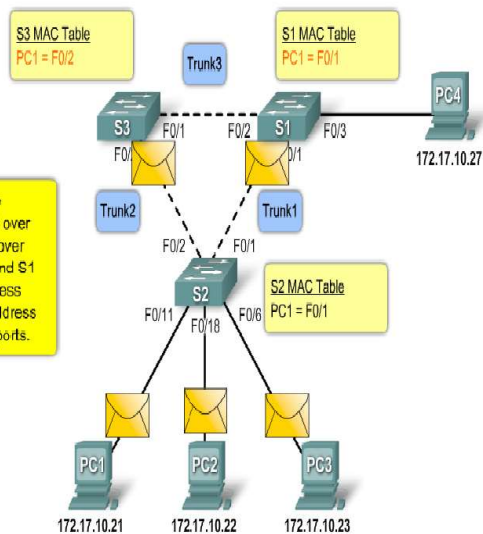
layer2 loop

PC1 sends a broadcast to switch S2. Switch S2 updates its MAC address table with PC1's MAC address on port F0/11.

Switch S3 forwards the broadcast to switch S1. Switch S1 forwards the broadcast to switch S3. Both switches update their MAC address with the new port on which it received the broadcast.

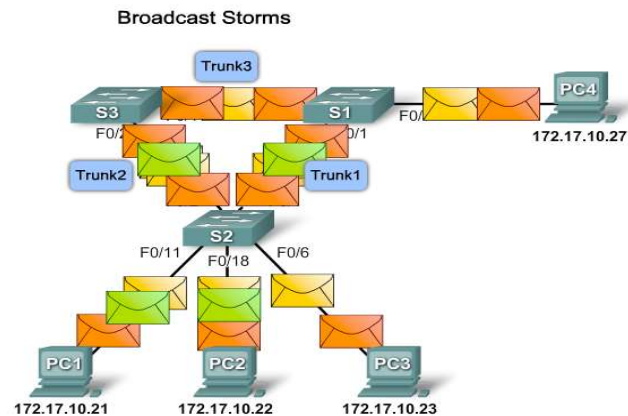
Switch S2 forwards the broadcast to switch S3 over Trunk2 and switch S1 over Trunk1. Switches S3 and S1 update their MAC address tables with the MAC address of PC1 on their Trunk ports.

Switch S1 forwards the broadcast frame out Trunks 1 and 2 to switches S3 and S1. Switches S3 and S1 update their MAC address table with the new entry for PC1.



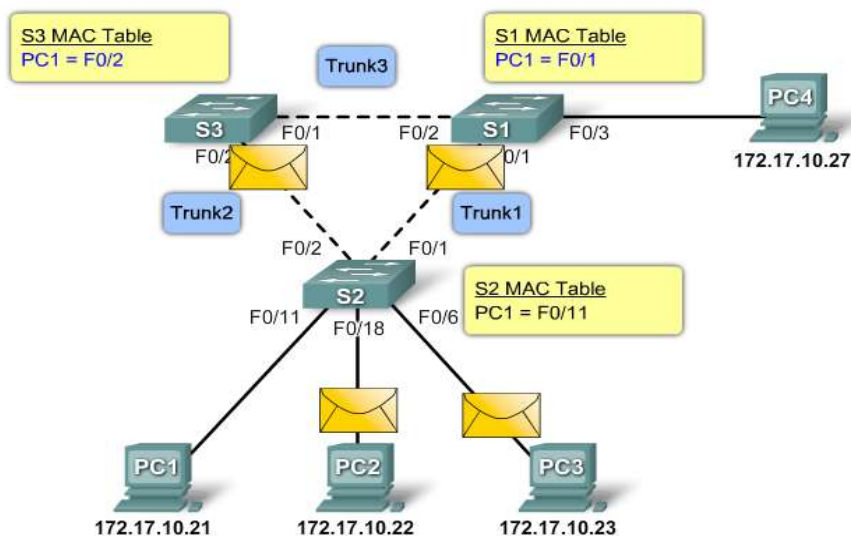
broadcast storm

A broadcast storm occurs when there are so many broadcast frames caught in a Layer 2 loop that all available bandwidth is consumed.



7

Duplicate Unicast frame



BPDU

BRIDGE PROTOCOL DATA UNIT

Bridge priority	MAC address
-----------------	-------------

32768

000A00222222

9

Choose the root bridge

- Each switch has a bridge ID (BID) of priority value followed by MAC address
- Switches exchange Bridge Protocol Data Units (BPDUs) to compare bridge IDs
- The switch with the lowest bridge ID becomes the root bridge
- Administrator can set the priority to fix the selection

10

Spanning tree algorithm

The switches use this algorithm to decide which ports should be shut down.

1. Choose one switch to be “root bridge”
2. Choose a “root port” on each other switch
3. Choose a “designated port” on each segment.
4. Close down all other ports.

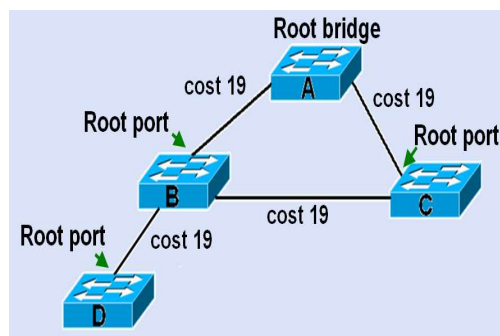
11

Select root ports

- Every non-root bridge (Switch) selects a root port
- This is the port with the lowest cost path to the root bridge

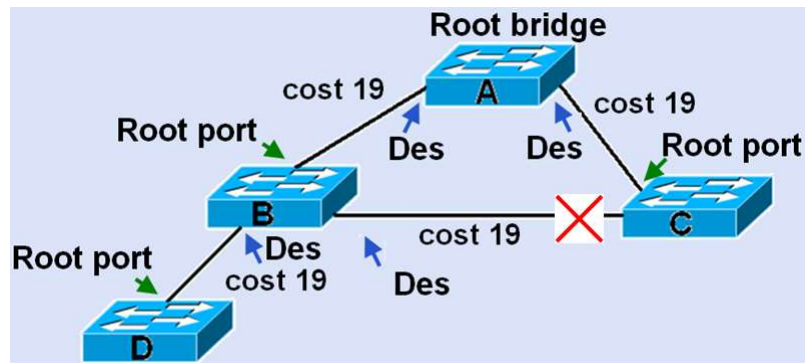
Link Speed	Cost (Revised IEEE Specification)
10 Gb/s	2
1 Gb/s	4
100 Mb/s	19
10 Mb/s	100

Set by IEEE



If the cost is same and Close down redundant links

Choose the port on the switch with the lower bridge ID.
Suppose this is switch B.



Best path

